Quantum

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1 Introduction and Motivation

Starting from a historical and philosophical optic of Management, I would end up in explaining why there is the research for a management that embraces uncertainty as a source of creativity, as stated in "Quantum Management"; theory created by Donah Zohar.
She arrived to this work after writing "The Quantum Self", here I would like to point out what is "Self" in Psychology, in particular, from a Jung's point of view, that saw the "Self" as a "Circle".
This geometric shape has been reshaped throughout the history by philosophers such as Hegel and Heidegger (hermeneutic circle). (Indeed, Danah Zohar studied also philosophy, with a focus on Heidegger.)
This would lead the path to a relation with cause-effect that is not anymore linear, indeed, it could be inspired by "synchronicity", topic dealt by Jung and Pauli, in their correspondence.
This creates a basement to talk about determinism and indeterminism.
De Felice notes + assignment
Why do we need quantum management, and how it is inspired by quantum mechanics and crisis of classical mechanics.

2 Classical Mechanics and Quantum Mechanics

Review of Mathematical Methods of Classical Mechanics (Hamiltonian and Lagrangian) adapted to very principal topic of Classical Mechanics such as kinematics, dynamics, thermodynamics, optics etc. And state which problems were unsolved such as catastrophe UV.

Switch to Quantum Mechanics similarly
Table to compare them, i would take shrodinger equation and newton equation.
End with Klein Gordon equationa and the introduction to "Relativistic Quantum Mechanics" and which is the difference from the non-relativistic case.

3 Determinism and Indeterminism

Subtle is the lord (Pais)
Inward Bound (Pais)
Can we know before it happens? Or as Einstein asks Pais "Do you really believe the moon is not there when you are not looking at it?".
Here, I would introduce the classical mechanics crisis that led to a quantum theoretical formalism of mechanics. Explaining exactly which phenomena were not clear.
I would talk about main key points of quantum mechanics and classical mechanics, showing major differences.
In particular, I would talk about the EPR paradox, and how this opens the door to the "conflict" between Determinism and Indeterminism. (philosophically there would be a lot, but i am not so sure that is a good choice talking about it here. In my opinion, it would be wiser pointing out philosophical, ontological and epistemological questions at the end.

here i would like to bring a more historical and factual chapter)

4 Newtonian Management and Quantum Management

Pros and Cons
p318, academic sciences, a table with Principles

5 Interpretations of Quantum Mechanics

Here, I would talk about some interpretations of quantum mechanics: agnostic, eretic and orthodox ("Quantum Mechanics at Crossroads"). Illustrating why the eretic/realistic interpretation could be maybe further investigated. Point out the Von Neumann theorem and Bell's considerations. Thus to have two major pillar between indeterminism (Copenhagen, density matrix, **Von Neumann**) and determinism (Bohm mechanics, - De Broglie pilot wave, Bell, Valentini, John Bush, Couder & Fort)

5.1 Quantum Management with the De Broglie-Bohm Interpretation of Quantum Mechanics - The case of the Pilot Wave

Analogy with a particle moving in electromagnetic fields (study that I previously made for an exam) and hydrodynamic pilot wave (Bush, Couder & Fort)
The role of analogy/Ex-post analogy: can we deduce a physical principle from an analogy?
Relativity, Space, Time and Light
Time diagram (QED, Feynman)

5.2 Quantum Management with the Copenhagen/Von-Neumann Interpretation - The case of the density matrix

Relativity, Space, Time and Light
Time diagram (QED, Feynman)

5.3 Von-Neumann Entropy, No Hidden Variables Theorem and Bell's Inequalities

6 Determinism (Pilot Wave) and Indeterminism (Density Matrix) in Management

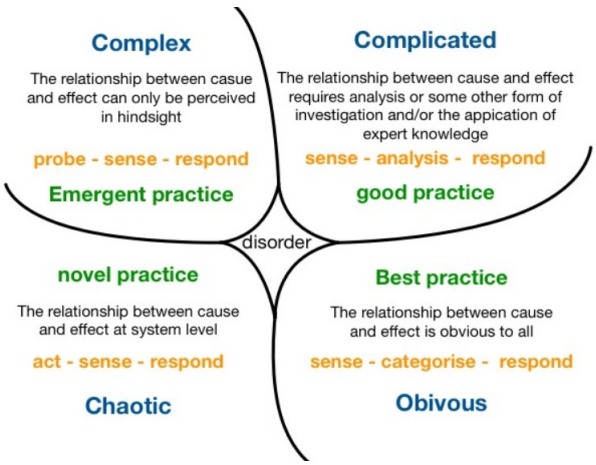

Figure 1: 5 Domains

Comparison between Copenhagen/Von Neumann interpretation and Bohm. interpretation.

7 Analogy between Quantum Mechanics and Insurance Management

7.1 Analogy between Quantum Mechanics and Quantum Leadership

7.1.1 Comparing Agile and Quantum Management

Agile management revolutionized project delivery by inverting the classical project management triangle and focusing on iterative, incremental value. However, Agile's mathematical foundation, while flexible, is still largely linear and deterministic. Quantum management takes this a step further, embracing a probabilistic and interconnected worldview. Below, we compare the mathematical underpinnings of Agile with a conceptual re-interpretation from a quantum perspective.

7.1.2 The Project Management Triangle

* **Agile Model:**

$$f(S_i) \approx T_{fixed} \times C_{fixed}$$

Here, the delivered scope (S_i) for an iteration is the primary variable, determined by fixed time (T_{fixed}) and cost (C_{fixed}) constraints.

* **Quantum Management Model:** From a quantum perspective, the project management triangle is in a state of **superposition**. The scope, time, and cost of a project are not single, fixed values but exist as a probabilistic combination of all possible outcomes until a specific "measurement" or decision is made.

7.1.3 Velocity and Sprint Planning

Velocity is a core metric for forecasting in Agile, but it's often treated as a deterministic average, which can be misleading in a complex environment.

* **Agile Velocity:**

$$V = \frac{1}{N}\sum_{i=1}^{N} S_i$$

This formula calculates the average velocity (V) based on historical scope completed (S_i) over N sprints.

7.1.4 Forecasting Project Completion

Agile forecasting uses a linear model, projecting a steady velocity into the future. This is a useful heuristic but fails to account for non-linear events.

* **Agile Forecasting:**

$$T_{sprints} = \frac{TotalWorkinBacklog}{Team'sVelocity}$$

$$D_{completion} = D_{current} + T_{sprints} \times T_{sprint}$$

7.1.5 Burndown/Burnup and The Observer Effect

Agile charts provide a visual representation of progress, but the very act of tracking them can alter the system's behavior.

* **Agile Charts:**

$$RemainingWork_t = TotalInitialWork - \sum_{i=1}^{t} WorkCompleted_i$$

This equation provides a seemingly objective measure of progress.

7.2 Analogy between Quantum Mechanics and ESG principles

7.2.1 Superposition and Uncertainty in ESG

- **Quantum Mechanics Concept:** A particle exists in all possible states until its state is measured.
- **ESG Principle:** A company's ESG performance is not a single, fixed reality. It exists in a state of **superposition**—simultaneously being a leader in some environmental metrics while also carrying significant social or governance risks. For example, a company might excel in renewable energy adoption (E_1) while simultaneously having an opaque and potentially unethical supply chain (S_2). The future of its ESG trajectory is not a predetermined outcome but a complex probability distribution.
- **Quantum Decision-Making Approach:** A quantum manager would resist the urge to simplify this reality into a single, definitive ESG score. Instead, they would embrace the inherent uncertainty and the full range of possible outcomes. The focus shifts from achieving a perfect score to building **resilience and adaptability**. This involves:

- **Scenario Planning:** Actively mapping out the most likely positive and negative ESG scenarios for the company.
- **Probabilistic Thinking:** Moving beyond binary thinking (e.g., "we are ESG-friendly") to assessing the probabilities of various outcomes.
- **Avoiding Premature Collapse:** Refraining from a rigid, single ESG strategy that "collapses the wave function" of possibilities, potentially ignoring crucial risks or opportunities. The goal is to remain agile and prepared for multiple eventualities.

7.2.2 Entanglement and Interconnectedness

- **Quantum Mechanics Concept:** Two particles can be so linked that a change in one instantly affects the other, regardless of distance.
- **ESG Principle:** Environmental, Social, and Governance factors are not independent silos; they are profoundly **entangled**. A company's carbon footprint (E) is inseparable from its labor practices (S) and its board's oversight (G). For instance, a decision to use a cheaper, less sustainable supply chain might superficially improve the bottom line and local environmental metrics, but it is deeply entangled with potential human rights abuses and lack of governance transparency.
- **Quantum Decision-Making Approach:** A quantum manager recognizes and works with these entanglements, understanding that optimizing for one factor in isolation is a futile and dangerous exercise. The approach is defined by:
  - **Holistic Strategy:** Designing an ESG strategy that considers the interconnectedness of all factors. This means a climate risk strategy must be considered alongside employee well-being and governance structures.
  - **Non-Linear Thinking:** Acknowledging that the relationship between ESG factors is not a simple cause and effect. A small change in a governance policy (e.g., a board mandate for ethical sourcing) could have massive, non-proportional impacts on social and environmental outcomes. This requires a systemic view rather than a linear, reductionist one.

7.2.3 The Observer Effect and Measurement

- **Quantum Mechanics Concept:** The act of measuring a system changes its state.
- **ESG Principle:** The metrics a company chooses to measure its ESG performance fundamentally change its behavior. Setting a specific Key Performance Indicator (KPI), such as a particular carbon reduction goal (E_{KPI}), can cause a "measurement collapse," where the organization focuses exclusively on hitting that metric, potentially at the expense of other important but unmeasured factors. The metric itself becomes the reality, often distorting true long-term impact.
- **Quantum Decision-Making Approach:** A quantum manager is acutely aware of the limitations of ESG metrics and reporting. Their philosophy is:
  - **Metrics are not Reality:** The numbers in an ESG report are an incomplete picture of a company's true impact. The focus should be on the underlying values and intentions, not just the reported figures.
  - **Measurement Creates a Response:** It is crucial to choose metrics that align with the company's true long-term values, not just what is easy to measure or what will garner the best short-term rating. This leads to a focus on more qualitative, holistic measures of impact and a culture of continuous learning and adaptation, where metrics are seen as tools, not as the ultimate goal.

7.2.4 The Uncertainty Principle and Trade-offs

- **Quantum Mechanics Concept:** It is impossible to know two complementary properties (e.g., position and momentum) with perfect precision simultaneously.
- **ESG Principle:** This concept directly applies to the inherent trade-offs in ESG. You cannot have perfect precision on all ESG goals at once. A company might face a trade-off between maximizing short-term returns for shareholders (G) and making a long-term, costly investment in an environmentally friendly technology (E). Trying to optimize for one factor with perfect certainty will inherently increase the uncertainty or diminish the performance of others.
- **Quantum Decision-Making Approach:** A quantum manager does not try to eliminate these trade-offs. Instead, they make them transparent and manage them deliberately. This approach involves:
  - **Transparent Trade-offs:** Clearly articulating the choices and their consequences to stakeholders. For example, explaining that a short-term dip in profit is a necessary trade-off for a significant long-term reduction in climate risk.

- **Navigating the ESG Landscape:** Recognizing that every decision is a move within a multi-dimensional "ESG landscape," where a gain in one dimension often comes at a cost in another. The goal is to find a dynamic balance, not a static, perfect state.

7.2.5 Quantum Decision-Making: The Density Matrix Framework

Quantum probability and quantum decision-making
Kahnemann-Prospect Analysis
VonNeumann- Behavioural Economics
The transition from Agile's linear equations to a quantum perspective provides a more realistic and nuanced view of management. This opens the door to powerful new tools for managing complexity. One such tool is the **density matrix theoretical framework**, which perfectly bridges the gap between quantum mechanics and the probabilistic nature of modern management.

The density matrix (ρ) is a mathematical object used in quantum mechanics to describe a system whose state is not known with certainty. It represents a statistical mixture of possible states, not just a single pure state. This framework is a perfect fit for fields like **insurance management**, where risk and uncertainty are the central challenges. This conceptual shift from Agile's deterministic math to the probabilistic framework of the density matrix offers a new way to understand and manage risk. It forces us to confront the inherent uncertainty of complex systems, moving beyond simple averages to a more nuanced and powerful approach.

7.3 Analogy between Quantum Mechanics and Market

Quantum Market Decisions
Quantum Probability based Decision Making in Finance: from Individual Preferences to Market Outcomes
Comparison of Markov versus quantum dynamical models of human decision making

8 Quantum Information Theory and Insurance Governance

Here, I would briefly describe the 2022 Nobel Prize "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science".

I would describe quantum information and quantum computing, with related quantum cybersecurity and quantum machine learning.

8.1 Quantum Computing in Portfolio Management

A quantum model for the stock market
Density matrix
Here, I would do a compilation work (risk-return) starting from the previous chapter about Market, but also a more hands on through an application with quantum computing and a neural network, possibly. It could also be just a begin or an introduction. If I do not have time, I could just describe by a mathematical point of view how this could work. Just, a pictorial view, at the moment (like a flow diagram with mathematical equations)

8.2 Quantum Computing in Risk Management

I have already done some work for De Felice's assignment, and I just have to develop it further (i can share with u). Indeed, there is a lot also about quantum credit risk analysis (i made an analysis with the Ising model for an assignment, i could further develop this, as well as what I described for Moriconi's assignment-this last i could implement it in R, as i did just a "descriptin").

8.3 Quantum Machine Learning for Insurance and the Inverse Problem

Here, i would do a hands on job, other than a compilation one.
I would focus on uncertainty and create an analogy between NNPDF/SIMUnet (that i used and applied for my masters thesis) and the prediction of risks in insurance, through the "inverse problem". (It would also be interesting to start introducing a quantum arrangement of NNPDF for risk-predictions, maybe just also mathematically, by writing equations, without still implementing a code).

8.4 Quantum Cybersecurity and Non Commutative Properties

Here, i made a study about quantum topological cybersecurity (i do not know if it still exists), i could just develop it further. i also studied the non-commutative property in topological qubits for cybersecurity goal!

8.5 Quantum IOT & CLOUD

i would take my assignment (that i did about ESG) and i would extend it to quantum esg principles (from the previous chapter about it). i think it does not have to be particularly rigorous.

8.6 Quantum Data Base System

Quantum Databases for Dynamic Data Storage, CERN
i would do hands-on and compilation work on quantum computing in data base system. But, i had an idea about "dense coding", which is also used at CERN. i think i could talk about it and make a connection with a customer centricity analysis.

9 Simulation of Quantum Management with a Neural Network

Here, i would introduce a new neural network that simulates the quantum management.
Again, if there is time to start implementing, great. Otherwise, there is the mathematics, and i think it's crucial to deal, as first place, with it, in order to simulate it.
Then, in a future, it would also be interesting to create a neural network with code.

Here, i would do a more "soft" insight about the relation between quantum management and the society it self. In my opinion, "quantum" is not only a mere practical application, but, if well handled, it is a "Mindset", that could be applied in every field. So, i think that it's a topic that could be easily described also from softer points of view, such as follows.

10 Art, Literature and Spirituality in Quantum Management

Piero dalla Francesca (prospettiva), yin-yang, Japan view, Antigone

11 Social Change and Psychology in Quantum Management

A well-tempered determinism, hysteresis in economics and in social change

12 Insurance Innovation: Quantum Self and Quantum Management

I would end with this principal chapter by describing how an analysis that embraces uncertainty and chaos, as quantum management would allow, would possibly lead to a better customer analysis, in a "centricity" optic. That, in my opinion, is the core of insurance innovation (citing Lemonade, for example).